3.4.1 Further Mechanics

• Momentum concepts

Force as the rate of change of momentum

$$F = \frac{\Delta(mv)}{\Delta t}$$

Impulse $F\Delta t = \Delta(mv)$

Significance of area under a force-time graph.

Principle of conservation of linear momentum applied to problems in one dimension.

Elastic and inelastic collisions; explosions.

• Circular motion

Motion in a circular path at constant speed implies there is an acceleration and requires a centripetal force.

Angular speed $\omega = \frac{v}{r} = 2\pi f$

Centripetal acceleration $a = \frac{v^2}{r} = \omega^2 r$

Centripetal force $F = \frac{mv^2}{r} = m\omega^2 r$

The derivation of $a = v^2/r$ will not be examined.

• Simple harmonic motion

Characteristic features of simple harmonic motion.

Condition for shm: $a = -(2\pi f)^2 x$

$$x = A\cos 2\pi ft \quad \text{and} \quad v = \pm 2\pi f\sqrt{A^2 - x^2}$$

Graphical representations linking x, v, a and t.

Velocity as gradient of displacement-time graph.

Maximum speed $= 2\pi fA$

Maximum acceleration $= (2\pi f)^2 A$

• Simple harmonic systems

Study of mass-spring system.

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Study of simple pendulum.

- ## Forced vibrations and resonance

  Qualitative treatment of free and forced vibrations.
  Resonance and the effects of damping on the sharpness of resonance.
  Phase difference between driver and driven displacements.
  Examples of these effects in mechanical systems and stationary wave situations.